SONOGRAPHY

REVIEW ARTICLE

The role of deliberate practice in development of essential sonography skills

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Abstract

The role of deliberate practice (DP) theory in the development of expert performance has been widely studied. This paper reviews the concept of DP and the various factors that could impact the development or refining of essential sonography skills for students and practitioners. In this article, the authors review the concept of DP for its potential benefits and limitations in Medical Sonography education and its potential impact on the future calibre of practitioners. The paper encourages both novice and experienced practitioners to rethink how DP could enhance the notion of grasping a skill. Evidence from various areas of practice including the health professions has been used to argue that sonographers can successfully incorporate DP. This review opens up the opportunity for further conversations on how various skills could be isolated and improved using DP theory, with the primary aim of improving the quality of patient care in sonography practice.

Introduction

Extensive research has shown that deliberate practice (DP) can influence the development of expert performance beyond any innate individual differences in various domains.[1–3] DP refers to focused, goal-oriented activities that are designed to improve a complex skill. During these activities, the learner concentrates fully, practices the skill deliberately and interprets feedback (directly or via a coach or mentor) with the aim of achieving a new higher level of performance.[4] To date, the literature does not support any single factor responsible for expert performance, but many examples, particularly in the health professions, support its use to accelerate skills development.[1,5]

All imaging modalities are influenced by the skill of the operator, for example, to position the patients correctly, to select the appropriate protocol relevant to the region of interest and to make appropriate adjustments to align with the unique characteristics of each patient. Sonography requires a number of simultaneous complex skills. Transducer manipulation in the various anatomical planes is one; in which the sonographer uses a multitude of combinations of pressure, angulation, rotation and tilt to produce a diagnostic image. Control of multiple parameters that are not automatically adjusted by the machine is another; these are fine-tuned to adjust for the manipulations, and for the manifold conditions encountered in patient anatomy and habitus, to attain an optimised image. Simultaneously, the sonographer also interprets the multiple images per second gained during acquisition to determine normality of the organs and structures encountered, recognising and working around or compensating for, the unique imaging artefacts encountered because of the physics of ultrasound. Where anomaly and/or pathology is present, decisions are made in light of clinical reasoning skills, incorporating the patients' clinical and imaging history and further information gained such as physical findings, during the scan. In other imaging modalities, an image may be rejected because of suboptimal technical quality, but the sonographer rejects the majority of the scan because they do not provide the diagnosis. The sonographer must make the diagnosis and be able to interpret the scan to document the presence of any abnormalities. The diagnostic ability of an ultrasound scan remains in real time, and at the point of care; if significant pathology is present and not imaged

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appropriately, it will not be detected in a review by a radiologist, certainly when only static images are available. As noted, radiologist and researcher Harris J. Finberg says, '… in large measure, the one who holds the transducer makes the diagnosis' (p. 1544).[6] This combination of abilities highlights the high level of skills required of a sonographer.[7] The unique link between the diagnostic quality of the ultrasound image and the inherent skill of the sonographer requires particular attention to the acquisition and refinement of both psychomotor skills and the cognitive aspects of clinical reasoning.

These attributes make sonography similar to many activities that require a combination of skills to perform well. For instance, playing a musical instrument or mastering a particular sport both require exceptional coordination, as well as deep knowledge of music or the game. DP has been applied successfully to the acquisition of psychomotor skills in these areas.[4,8] More recently, there have been compelling arguments for the adoption of DP theory to facilitate skills acquisition in healthcare professions such as Nursing and Medical practice.[1,2,9] Eppich *et al.*[10] postulate that the goal is to make learning deliberate, clear, and objective with ample time allowed for skill progression. Despite this adoption in other health disciplines, there is a paucity of literature that indicates how DP theory can be utilised in sonography practice.

This paper aims to review the concept of DP and discuss how it can be applied to refine essential sonography psychomotor skills to foster competency and confidence in practitioners.

Approaches to teaching and learning

Sonography education providers traditionally focus their attention on core knowledge and assessment of competency to align with accreditation standards.[11] Attention should be further focused on enhancing a student's skills while in the clinical environment. While the ultimate responsibility for learning lies with the student,[3] the curriculum ought to provide an efficient framework that scaffolds learning to develop cognitive and psychomotor skills. Additionally, sonographer education inevitably goes beyond the time frame required to complete a university qualification or an accredited course. Like other clinically oriented professions, ongoing skill development occurs through observation, hands-on clinical practice and a concrete feedback mechanism from educators or mentors. DP-based psychomotor skill acquisition should occur primarily within the clinical environment where students are engaged in authentic tasks of performing ultrasound examinations using clinical quality machines and engaging with patients and other health practitioners. Contemporary evidence also attests to the benefit of technologies that simulate clinical processes in educational environments, and the effectiveness of simulation as a pedagogical approach has also been demonstrated for sonography students.[12,13]

The ultimate aim is to engage students in learning opportunities that effectively and safely prepare them for clinical practice.[13] Additionally, to provide new graduates with a deep understanding of the theories of effective skills development to foster life-long learning. We argue here that DP theory could become an essential tool to engage both students and practitioners in activities that optimise both psychomotor and cognitive skills.

The history and application of deliberate practice

Deliberate practice was first developed by Anders Ericsson, professor of psychology at Florida State University, who researched human performance in various domains. He and colleagues provided the first empirical evidence for DP, describing the process by which musicians developed expertise by engaging in focused practice activities with individualised feedback developed by their teachers, finding that individual differences amongst elite performers are closely related to the accumulated DP over many years.[4] Studies were also conducted in other domains such as chess and typing proficiency demonstrating improved performance was a function of more experience when coupled with DP.[14–16] From this foundational research and other research into working memory,[17,18] Anders Ericsson coined the term Deliberate Practice to describe a method designed to foster expert performance.[2,4] Several iterations have since been provided in the literature, and many authors agree that DP should include; repetitive performance of cognitive and motor skills, a formal feedback mechanism to allow for correction, and additional opportunities for practice until a satisfactory or expert level is attained.[3,19,20] The book 'Peak – Secrets from the new science of expertise' provides a more comprehensive definition of DP and draws a clear distinction between it and purposeful practice; DP 'is both purposeful and informed and guided by … an understanding of what … expert performers do to excel. Deliberate practice is a purposeful practice that knows where it is going and how to get there' (p. 98).[21]

Sperry[22] suggests that DP should be 'intentional' and be aimed at achieving a level of expertise that is beyond an individual's level of proficiency. In the context of medical education, McGaghie and Kristopaitis[9] provided a much broader definition of DP as a concept where learners are 'engaged in difficult, goal-orientated work, supervised by teachers, who provide feedback and

correction, under conditions of high achievement expectations, with revision and improvement'. Harris *et al.*[23] simply defined DP as a means by which human performance can be iteratively improved, adding an effective coach who gives the student the best chance to improve more quickly than simply spending time 'just doing'. Ericsson *et al.*[4] also make the point that this type of practice is designed to force adaptations in which each learner will develop their own novel way to solve a complex issue. Ericsson and Pool[21] described these as 'mental representations', the strength and quality of which helps define the novice or beginner practitioner from those that are truly expert in their field. These authors provided an example of how expert surgeons explained their thinking to students during complex operations to tease out the mental representations, which may, in fact, be quite subconscious, but made a lasting difference to the students learning. Understanding these, and other strategies consistent with DP such as peer evaluation and using 'in time' and micro debriefing strategies[1–3,5,20,24–27] are used effectively in medical education, all of which have great potential for sonography.

Stages of learning and continuing professional development

There is the possibility of assuming that once a practitioner has finished training and gained their qualification that they are suitably skilled in performing at a high or expert level. Ericsson and Pool[21] argue that this is far from the truth, and this type of institutional thinking can be detrimental to practice, as without repeated practice or continual use of skill, drift will occur from the baseline. A meta-analysis by Choudhry *et al.*[28] found that the care doctors provide over time grew worse, or at best stayed the same, as the number of years of practice increased. After a certain point in time, some practitioners may become comfortable with their abilities and therefore without knowing, fail to progress. Gonzalez and Kardong-Edgren[25] add that prolonged periods of non-use of certain skills can lead to loss of mastery and skill. Ericsson *et al.*[8] also contend that the more one engages in optimal DP, the better they are likely to become.

The Australasian Sonographers Association (ASA) competency framework cites the Dreyfus model of skill acquisition,[29] in which the five stages of learning are aligned with the level of competency and the practice pathways (Table 1). Field[30] demonstrates how the Dreyfus model of skill acquisition can be mapped to the progression from novice to expert and master in sonography. Both Field[30] and the ASA place basic training at the 'competent' level. The ASA reminds practitioners of the role of continuing professional development (CPD), providing advancement to the level of proficiency and a pathway towards expert practice. As the DP literature has demonstrated, continued improvement of skills will only occur when the key components of learning and practice are adhered to, repetition of practice is maintained, and feedback is provided.

Most healthcare professions are excellent at promoting and providing CPD opportunities. Many practitioners attend conferences regularly to improve their knowledge in various discipline-specific areas. Some authors caution that attending conferences and lectures can be a futile effort if the aim is to make a lasting change to areas of practice that involve a complex skill.[2,31] However, when the aim is to improve a specific shortcoming in practice performance, CPD activities that involve workshops, where a practitioner can gain hands-on training while being instructed by an expert, gaining feedback on how to improve in a particular area, are most effective.[32]

Deliberate practice in other fields

Several authors have explored DP utilisation in the health professions, including McGaghie and Kristopaitis[9] and McGaghie *et al.*[26] who proposed that DP be used as a core component of medical education and as it promotes frequent monitoring and error correction and leads to better health for individuals and populations. Other researchers have found DP useful in nursing education,[19,20] for example, Ross *et al.*[33] successfully implemented DP coupled with peer mentoring, which

Table 1 The five stages of learning and sonography practice pathways

Dreyfus learning stages	Sonographer	Pathways
1. Novice	Student	University/Professional qualification
2. Advanced beginner	Student	University/Professional qualification
3. Competent	Entry level sonographer	Graduate
4. Proficient	Experienced practitioner	Continuing professional development
5. Expert	Advanced practitioner	Further study

Adapted from Australasian Sonographers Association Competency Standards for the entry level sonographer, October 2009, p. 4.

enhanced students' mastery and retention of essential skills. Gonzalez and Kardong-Edgren[25] identified DP as beneficial to ensure competency and retention in the nursing profession. Some advantages include gaining economy of movement, increasing speed and reducing variability across a range of tasks.[34,35]

Campitelli and Gobet[36] found DP to be a necessary condition to achieve high levels of performance in chess. However, they also argued that DP alone is not sufficient. One condition highlighted in the literature is that an inherent love or enjoyment of the activities is vital to reach an expert level. Without such inherit motivation, it is unlikely that any practitioner will truly engage in the many hours of practice and fine-tuning required to excel. Therefore, this implies that the implementation of DP alone in sonography or other health professions also requires that researchers explore other variables related to expertise.

Many authors acknowledge that every profession is different and how each adopts the concept of DP is unique.[37] It is important to explore what we can learn from other fields. Therefore, this review encourages to open the opportunity for a conversation as to how psychomotor skills can be acquired more efficiently and whether DP could be a useful mechanism to support this goal in sonography education.

Deliberate practice implementation in sonography

The implementation of DP in sonography can be investigated at different levels. As highlighted, DP occurs when an individual intentionally repeats an activity to improve performance.[38] At an individual level, a practitioner who is keen to improve his or her expertise can utilise the DP concept. However, the individual must identify a coach who provides immediate feedback and support. DP could be implemented in clinical practice environments particularly in a facility with skilled practitioners who can mentor other practitioners, utilising the DP approach. Academic and clinical institutions should be at the forefront of engaging in DP research activities. Most importantly, sharing of the strategies and outcomes in the sonography community is critical to the success of DP, and how it could be applied most effectively.

To further clarify the definition and general understanding of the DP concept, we offer a conceptual framework. Figure 1 shows a framework generated in light of the DP definitions and theory reviewed and our extensive knowledge of the field of sonography. However, it is pertinent that as evidence emerges from sonography-oriented DP approaches, empirical data and systematic reviews be used to modify and adapt this to the profession.

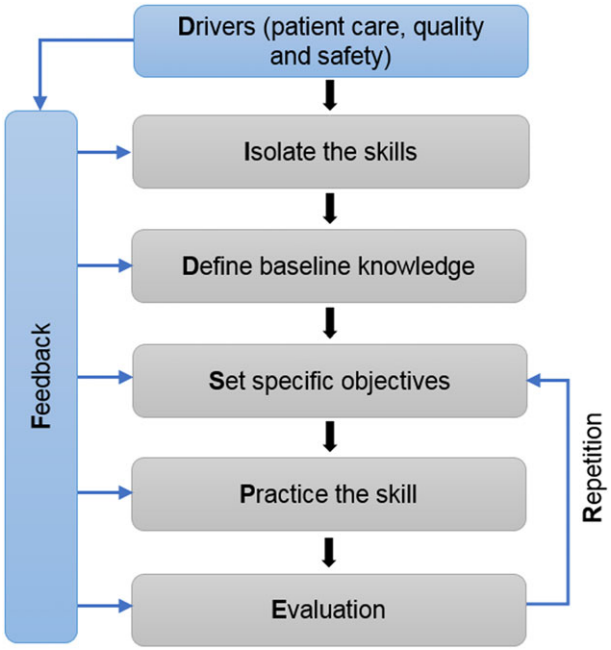

Figure 1 Proposed conceptual framework for deliberate practice in sonography.

Summary of the key action steps

Drivers: The practitioner identifies the need to improve, and in healthcare, it is important that the primary motivation for practitioners to accelerate a skill emanates from the need to improve quality of patient care and safety:

A1. *Isolate the skills* – Specific skills that need improvement are identified.
A2. *Define baseline knowledge* – Practitioners or a supervisor should determine baseline knowledge depending on the developmental level of the sonographer.
A3. *Setting specific objectives* – Clearly defined attainable goals should be documented that align with the identified skills.
A4. *Practice the skill* – Practice must be focused, systematic and carried out over a period that is sufficient to improve the skill.
A5. *Evaluation* – Evaluation is necessary to reach a mastery standard.[23]

Repetitive engagement: DP involves repetitively practising specific skills until they become routine.[37]

Feedback: DP approaches require feedback throughout the skill development process.

Further evidence to validate the importance of feedback from a skilled mentor, coach or clinical supervisor is provided in the succeeding texts.

The role of feedback and coaching

Highly skilled and operator-dependent tasks require additional elements to achieve a high level of performance. In DP, there is an emphasis on repeated activities, but a feedback mechanism with correction is crucial.[25] According to Ericsson and Pool,[21] the core components of DP include setting specific, attainable goals, focusing on the problem at hand and gaining feedback. Feedback helps practitioners correct and fine-tune their abilities and to obtain a higher level of skill. Boud and Molloy[39] further argue that the person receiving the feedback should be an active agent in managing and working with it. Eppich *et al.*[10] highlight the importance of feedback from a skilled coach, to support the learner to meet expert performance standards. In clinical settings, supervisors play a central role in the provision of feedback. However, it is important to clarify that, for feedback to be effective, it must be immediate, constructive and consistent, to allow for one to reflect on a particular skill. Rousmaniere[37] argues that although feedback is necessary for improvement, it is not itself sufficient. Research can determine those elements of feedback relevant to sonography professionals. In sonography, feedback is crucial particularly in the areas of pathology detection and the analysis of scan findings. As Gibbs[13] postulates, becoming an effective sonographer demands expertise in psychomotor competencies with regular practice complemented by expert feedback.

Ericsson[2] also identifies feedback as an important element in DP. He uses an example that in certain surgical fields, there is an almost linear improvement of skills over time but only when direct feedback is available. He uses the specific example of surgeons who perform prostatectomies for the treatment of prostate cancer. For the surgery to be effective, the entire prostate must be removed with histologically clear margins. A successful surgery has a direct impact on the patient's long-term survival rate, and the surgeon receives the results of their surgery a short time after they have performed it. It is clear that those surgeons who perform more operations have higher rates of patient survival than those who perform fewer surgeries. The more experienced surgeons have practised more, gained feedback and then adapted their technique to result in a better outcome.

Potential challenges

The major barrier to adopting DP in sonography could be lack of knowledge about the existence of DP theory and its application to skills development. Other challenges highlighted in the literature are the potential resistance to change or to try something new as well as the time to implement DP programs.[10] The repetition and focus that characterises DP is not an easy process; it requires time and energy.[37] It is difficult to isolate one particular aspect of a scan and allow the practitioner to spend sufficient time mastering it before moving to another. Without empirical evidence that links DP to improvement in skill, it may be difficult to convince sonographers to engage in DP approaches. However, implementing DP activities within a university environment and promoting its effectiveness to students may provide a catalyst to extend these types of activities to the clinical world beyond qualification.

There is also a need to distinguish work, or performance, and DP. According to Ericsson *et al.*,[4] work activities offer some opportunities for learning, but these opportunities are restricted because of the focus on performance and outcome. For DP to be effective, the focus must remain on the skill itself and the aspect of the skill that may require improvement. This type of self-monitoring, in which the practitioner is focused on spotting mistakes and adjusting accordingly, may be quite separate to the work required.

Conclusion

Deliberate practice has been shown to be a worthwhile theory in the refinement of psychomotor skills and the development of increasing levels of expertise. Evidence from many sources including healthcare professions suggests that with deliberate and repetitive practice, expert performance can be achieved. Feedback from a mentor has been identified as a critical factor in DP approaches. The benefits of DP, highlighted in our discussion, suggest that it could be worthwhile to embrace DP in sonography. However, there is a need to gather empirical evidence for DP in sonography practice and build evidence-based DP approaches that focus on refining essential sonography skills. We have proposed a conceptual framework to aid educators and/or practitioners to develop practical applications. Further research is needed in this potentially fruitful area.

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